Q-Cluster: Quantum Computing Simulator

Driving the future of quantum innovation through accelerated learning, advanced algorithm development and reliable quantum processor performance modelling.

Q-Cluster is a quantum simulation platform built to run over hybrid CPU–GPU computing infrastructure, covering the complete spectrum of Quantum Information Processing, from foundational quantum-algorithm learning through to modelling how quantum states evolve under noise. With configurable noise density matrices, it becomes a high-fidelity approximator for Noisy Intermediate-Scale Quantum (NISQ)-era behaviour.

A quantum computing laboratory without a dilution refrigerator
Algorithms that fail the way real hardware fails.

Tailored for educators, learners and researchers, Q-Cluster supports both ideal state-vector simulation and noise-modelled execution, letting users go beyond ideal-processor simulation to understand algorithm performance on realistic, imperfect quantum systems.

With an open, scalable architecture, Q-Cluster scales from 16 qubits on basic server-class compute up to 35 qubits on hybrid CPU–GPU HPC systems, with an easy-to-use GUI and on-premise access for students and faculty. Its noise-model integration supports learning in quantum-processor design and lets teams approximate and test noise effects before deploying applications on expensive quantum processors.

Key Features

- **GUI for Quantum Learning:** An intuitive graphical interface for educators, learners and researchers to build, run and understand quantum algorithms.
- **Noise Modelling:** Noise-integrated approximate simulations supporting conceptual and practical learning of quantum processor behaviour.
- **Algorithm Development and Application Prototyping:** Build and refine quantum applications, test robustness under ideal and noisy conditions, and approximate QPU performance scenarios.
- **Open and Scalable Simulation Infrastructure:** Scale from basic teaching modules to deep, large-scale quantum algorithms on a hybrid HPC backend.

Technical Specifications

Scalable Computing Infrastructure	CPU and GPU (Server / T4 / V100 / A100 / H100)
Simulation Type	Statevector, noisy density matrix
Simulation Capacity & Modes	Up to 35 qubits (statevector simulation) Up to 9 qubits (noise-enabled simulation)
Noise Models	Gate errors, decoherence, readout noise
Educational Mode	Circuit design tasks, quantum algorithm tutorials, guided modules for approximation of QPU behaviour
Supported Algorithms	Complete spectrum: Grover's, QFT, Shor's, Simon's, Deutsch–Jozsa and many more
QEC Support	3-, 5- and 7-qubit codes, encoding and decoding, and more
User Interface	Built-in drag-and-drop graphical interface for circuit design, execution and visualisation
Output	Probability graph distribution, basis state probability distribution, qubit representation on the Bloch sphere, and intuitive interpretation of quantum algorithms
Multi-user Environment	Classroom and laboratory deployment ready, JupyterHub compatible
Access	Web-based platform, accessed through common browsers including Google Chrome and Safari

No optical table, cleanroom or laboratory infrastructure is required for this platform.

Learning Outcomes

Students will learn:

- **Quantum Algorithms:** Implement foundational quantum algorithms on scalable qubit systems using guided modules and circuit-building interfaces.
- **Simulation of Noise Effects:** Build intuition on how gate errors, decoherence and readout noise affect quantum computations.
- **Quantum Error Correction:** Simulate and analyse logical qubit encoding and decoding schemes, and logical gate fidelity under configurable noise models.
- **Complete Spectrum QIP:** Investigate quantum algorithms from execution to design and evaluation, across ideal and noisy environments.
- **Quantum Application Prototyping:** Build, test and debug quantum workflows at up to 35 qubits noiseless and 9 qubits with noise, on hybrid CPU–GPU backends.
- **QPU Performance Approximation:** Approximate realistic quantum hardware behaviour under configurable noise density matrices for NISQ-era research.
- **Resource-Conscious Algorithm Scaling:** Understand how CPU and GPU infrastructure affects scaling, depth and execution limits.

Skills Gained

Students will gain:

- Quantum programming and algorithm development
- Noise-aware algorithm design
- Quantum error correction
- Quantum simulation
- Quantum algorithm optimisation
- Quantum application development
- Quantum hardware understanding

Industry Mapping

- Quantum software industry
- Finance and fintech
- Pharmaceuticals and chemistry (VQE)
- Logistics and transportation
- Healthcare
- Aerospace and defence
- Research

Current industry hiring analysis names finance, pharmaceuticals and materials science, aerospace and defence, and government and national-security programmes as the leading sectors recruiting for quantum-computing skills — with quantum algorithm design flagged as the single scarcest skill in the field, and Python-style programming plus noise-aware, NISQ-era simulation experience called out as consistently valuable across employers.

The platform enables students to progress from quantum fundamentals to practical algorithm development, simulation, quantum machine learning, optimisation and hardware-aware execution — skills increasingly relevant as organisations move from quantum experimentation toward application development. This enables institutions to develop industry-ready quantum talent capable of contributing to emerging quantum software, AI and ML, optimisation, scientific computing and quantum hardware ecosystems.

Curriculum Mapping

How Q-Cluster Enables the Key Learning Outcomes	AICTE Course Title (Course Code)
Basics of quantum computation; running quantum protocols in a quantum simulator; implementing simple quantum algorithms; simulation using statevector and density matrix approaches; noise modelling and system analysis based on QPU approximation; experimentation from basic quantum circuits to advanced research-level algorithms	Basic Laboratory Course for Quantum Technologies (QT04) Introduction to Quantum Computation (QT05)
The basics of programming — Python, Julia and QASM — using toolboxes for quantum technologies in these languages, and frameworks such as Cirq, PennyLane and Qiskit	Programming for Quantum Technologies (QTM 05)
The basics of qubits and gates; density operators and time evolution for mixed states; the working of quantum algorithms; basics of quantum error correction	Introduction to Quantum Computation (QTM 10)
Variational quantum algorithms, quantum-assisted machine learning, and applications of both	Advanced Quantum Algorithms (Elective)

QTXX refers to the AICTE Minor Degree for UG programmes in Quantum Technologies; QTMXX refers to the AICTE M.Tech Programme in Quantum Technologies.

Pre-requisites

- Basic Python programming
- Linear algebra and probability
- Basic familiarity with quantum computing concepts, helpful but not required, as foundational concepts are introduced within Q-Cluster's guided modules

Licensing Model

- **Software Licence:** Applicable when the software is deployed on a dedicated, in-house server for use by multiple users or systems.
- **User Licence:** Applicable when the software is deployed on an individual system, particularly in an AI laboratory where a single workstation is equipped with a GPU for dedicated use.

Requirements

Server deployment

- GPU server such as T4, V100, A100 or H100
- Computer laboratory

Individual deployment

- Computer laboratory, given that the computers have NVIDIA GPU.

Frequently Asked Questions

What computer, operating system or software is needed to operate it?

Q-Cluster is a web-based platform that can be accessed through commonly used browsers such as Google Chrome and Safari.

What is the licensing model like single-seat or lab-wide?

Both. Q-Cluster is designed to support concurrent student usage during laboratory sessions when deployed on a university server. A single-seat licensing model is offered for AI laboratories equipped with an individual server or workstation.

Is this cloud-hosted or on-premises?

Q-Cluster is currently offered to universities as an on-premise solution, deployed within the university's own infrastructure.

How many concurrent student seats does one licence cover?

The number of students who can use Q-Cluster simultaneously depends on the university's available server capacity and configuration.

What is the warranty period, and is an AMC or support plan available?

The warranty period is one year. An AMC and support plan is available for five years.

Is on-site installation and faculty or student training provided?

Yes. Q-Cluster is installed and configured on the university's premises. One training session is provided to faculty members during installation, to familiarise them with the platform and its key functionalities.

Is the software suitable for UG, PG or PhD-level work?

Yes. Guided modules for foundational algorithms such as Grover's and the QFT suit undergraduate teaching, while noise-modelled VQE and QAOA work and quantum error correction analysis are genuine postgraduate and research-level tasks.

What is the qubit-count difference between the noiseless (35) and noise-enabled (9) modes, and why?

Noiseless statevector simulation scales more efficiently than noise-modelled simulation. Modelling gate errors, decoherence and readout noise requires tracking a much larger density-matrix state space per additional qubit, which is why the noise-enabled ceiling of 9 qubits is substantially lower than the noiseless ceiling of 35 qubits on the same compute infrastructure.

Can experiments or algorithms be customised or extended by faculty?

Yes. Q-Cluster's architecture is explicitly open and scalable, designed for building and refining custom quantum applications rather than only running a fixed catalogue.